



AWS Academy Data Engineering
Welcome to AWS Academy Data Engineering
Student Guide
Version 1.0.6
200-ACDENG-10-EN-SG

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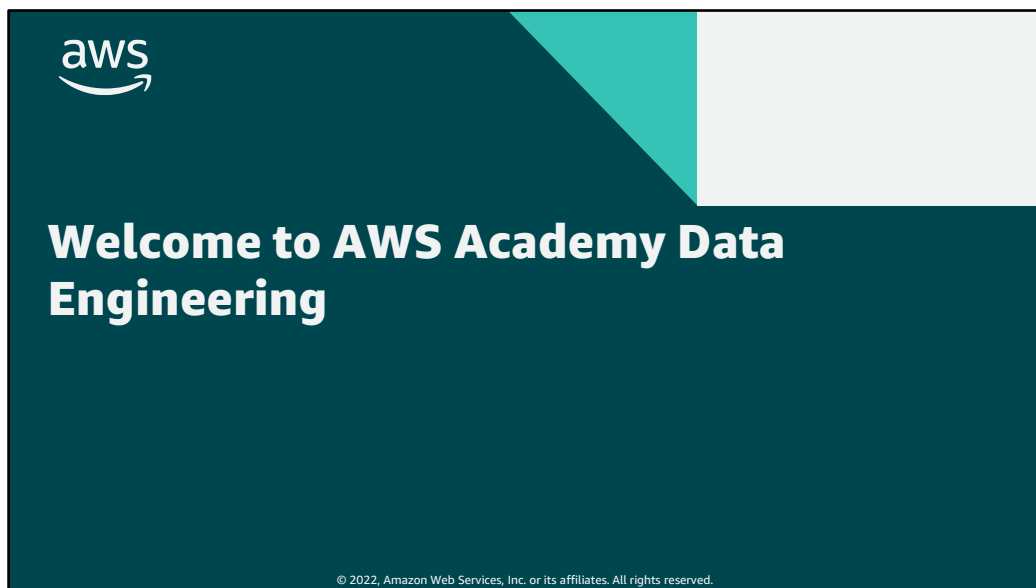
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Welcome to the AWS Academy Data Engineering course.

Module overview

Sections

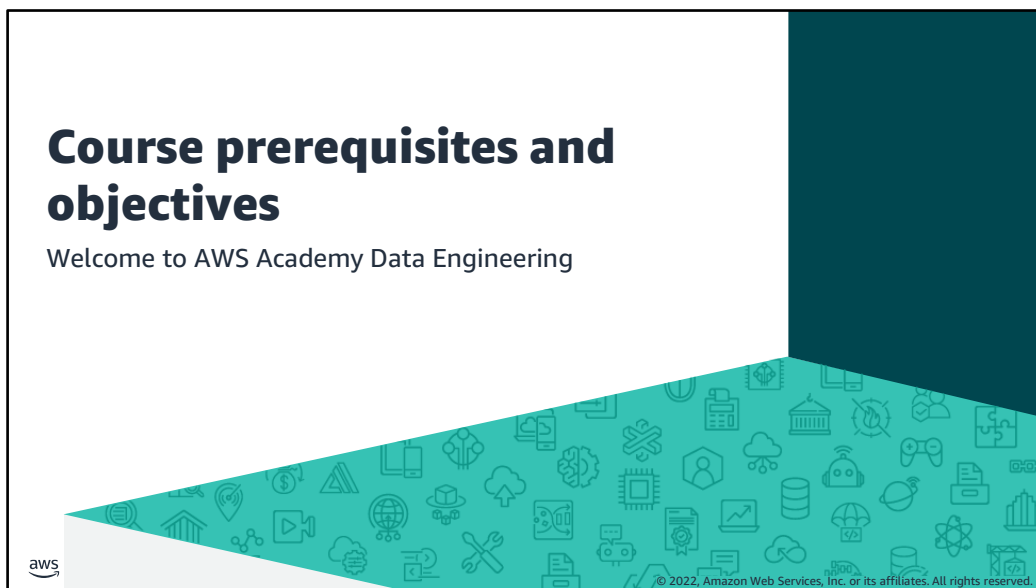
- Course prerequisites and objectives
- Course overview



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This module introduces you to the course.



This section provides information about the prerequisites and objectives of this course.

Prerequisites

- Completed the AWS Academy Cloud Foundations course or have equivalent experience
- Worked with Structured Query Language (SQL)
- Worked with databases
- Introduced to general networking concepts
- Understanding of decision-making knowledge in math, probability, and statistics



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This slide provides an overview of the knowledge and skills that you are expected to have prior to taking this course.

Related job roles

- Data engineer
- Data analyst
- Data scientist
- Extract, transform, and load (ETL) developer
- Machine learning (ML) practitioners



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This course is most aligned to a data engineer role. However, this course would also be appropriate for data analysts; data scientists; extract, transform, and load (ETL) developers; or machine learning (ML) practitioners who want to understand how the data that they use in their analysis and predictions is prepared for analysis using AWS.

Course objectives (Slide 1 of 2)

This course prepares you to do the following:

- Summarize the role and value of data science in a data-driven organization.
- Recognize how the elements of data influence decisions about the infrastructure of a data pipeline.
- Illustrate a data pipeline by using AWS services to meet a generalized use case.
- Identify the risks and approaches to secure and govern data at each step and each transition of the data pipeline.
- Identify scaling considerations and best practices for building pipelines that handle large-scale datasets.
- Design and build a data collection process while considering constraints such as scalability, cost, fault tolerance, and latency.



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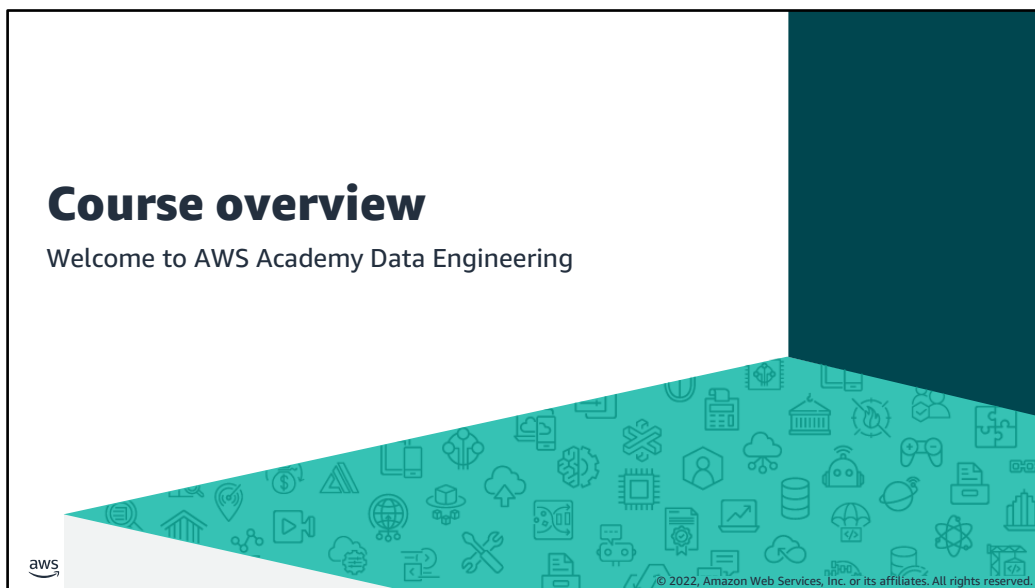
Course objectives (Slide 2 of 2)

- Select a data storage option that matches the requirements and constraints of a given data analytics use case.
- Implement the steps to process both structured, semistructured, and unstructured data formats in a data pipeline that is built with AWS.
- Explain the concept of MapReduce and how Amazon EMR is used in big data pipelines.
- Differentiate the characteristics of an ML pipeline and its specific processing steps.
- Analyze data by using AWS tools that are appropriate to a given use case.
- Implement a data visualization solution that is aligned to an audience and data type.



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This section provides an overview of the modules and learning modalities in this course.

Course outline

- Module 1: Welcome to AWS Academy Data Engineering
- Module 2: Data-Driven Organizations
- Module 3: The Elements of Data
- Module 4: Design Principles and Patterns for Data Pipelines
- Module 5: Securing and Scaling the Data Pipeline
- Module 6: Ingesting and Preparing Data
- Module 7: Ingesting by Batch or by Stream
- Module 8: Storing and Organizing Data
- Module 9: Processing Big Data
- Module 10: Processing Data for ML
- Module 11: Analyzing and Visualizing Data
- Module 12: Automating the Pipeline
- Module 13: Bridging to Certification
- IoT Use Case Slides



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To achieve the course objectives, the course consists of the modules that are listed on the slide.

The course also includes a set of slides called the IoT Use Case Slides. Your educator might use them to discuss an example of a data pipeline to ingest and process Internet of Things (IoT) data.

Course modalities (Slide 1 of 2)

Slides

- All modules

Activities

- Module 3: Planning Your Pipeline
- Module 4: Using the Well-Architected Framework
- Module 10: Labeling with SageMaker Ground Truth

Demonstrations

- Module 10:
 - Preparing Data and Training a Model with SageMaker
 - Preparing Data and Training a Model with SageMaker Canvas
 - Creating a Linear Regression Model Using Amazon Q Developer



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This course is presented by using a combination of slides, activities, demonstrations, and hands-on labs. Slides are the primary delivery method and are available for every module. The remaining modalities are distributed as shown on the slide.

Course modalities (Slide 2 of 2)

Labs

- Module 4: Querying Data by Using Athena
- Module 7: Performing ETL on a Dataset by Using AWS Glue
- Module 8: Storing and Analyzing Data by Using Amazon Redshift
- Module 9:
 - Processing Logs by Using Amazon EMR
 - Updating Dynamic Data in Place
- Module 11: Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards
- Module 12: Building and Orchestrating ETL Pipelines by Using Athena and Step Functions

Capstone Project



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This course is presented by using a combination of slides, activities, demonstrations, and hands-on labs. Slides are the primary delivery method and are available for every module. The remaining modalities are distributed as shown on the slide.

The Capstone Project provides an integrative project-based learning experience that reinforces technical skills that are taught in this course. The capstone offers you an opportunity to demonstrate critical thinking, problem solving, the software development lifecycle, and communication skills.

Learning assessments

Knowledge checks

- Provided for modules 2 through 12
- Drawn from the slides and student guide notes

Course assessment

- 25 randomized questions
- Drawn from the slides and student guide notes



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You can assess your progress throughout this course by completing the knowledge checks and the course assessment that are included with the course materials. Each module (except the first and last modules) includes a knowledge check with questions that are specific to that module. The course assessment includes questions from all the modules to test your overall course knowledge.

Both the knowledge checks and the course assessment questions are drawn from material found in the slides and student guide notes.

Preparing for certification

- This course is not intended to fully prepare you for related AWS Certifications.
- Review the Exam Guides for information regarding the certifications.
- The Bridging to Certification module provides information about additional exam preparation resources.

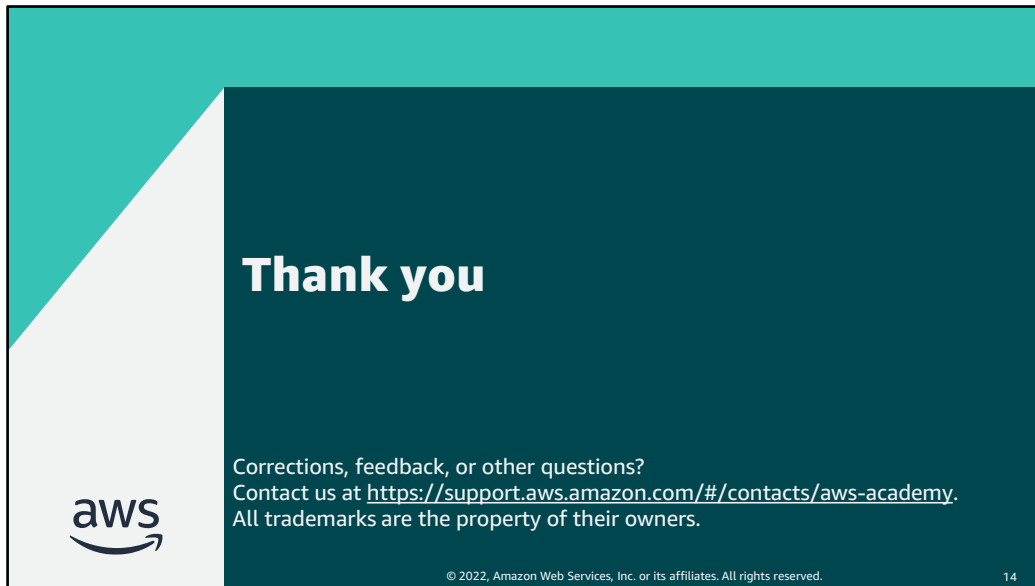


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While this course will introduce you to many aspects of data analytics and machine learning, the course will not fully prepare you for the related AWS Certification exams.

For additional information regarding the certifications, see the Exam Guides. Links are available in the Content Resources section of the course.



Thank you for completing this module.